Notes: Hornbeck & Rotemberg (JPE, 2024)
Growth Off the Rails: Aggregate Productivity Growth in Distorted Economies

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1 Big picture

- **Question.** How large were the aggregate economic gains from the nineteenth-century U.S. railroad network once we allow for distortions in the rest of the economy?
- **Core answer.** The railroads did much more than save transportation costs or raise land values. By increasing market access, they expanded production in counties where inputs were inefficiently underused, raising aggregate productivity through allocative efficiency.
- **Main empirical result.** A one-standard-deviation larger increase in county market access from 1860 to 1880 raises county productivity by about 20 percent. The effect is driven mostly by allocative efficiency (AE), not by total factor productivity revenue growth (TFPR).
- **Main counterfactual result.** Removing the 1890 railroad network would reduce U.S. aggregate productivity by 26.7 percent if worker utility is held fixed and population can adjust, or by 5.5 percent if total population is held fixed.
- **Mechanism.** Railroads lower trade costs, raising input use in connected counties. When the value marginal product of inputs exceeds marginal costs, expanding input use creates surplus greater than input costs.
- **Key distinction.** The paper does not argue that railroads directly reduce distortions. Instead, railroads increase activity in a distorted economy, and the expansion itself raises aggregate productivity.
- **Why previous estimates were lower.** Fogel-style social savings and land-value approaches assume or imply that secondary sectors are efficient. This paper shows that when other sectors are distorted, infrastructure can generate large indirect gains outside the transportation sector.
- **Policy intuition.** Infrastructure and general-purpose technologies can have especially large aggregate effects when they increase activity in sectors or locations with positive wedges between marginal products and marginal costs.

2 Incremental contribution of the paper

- **Beyond social savings.** The paper extends the classic railroads literature by showing that the value of railroads is not bounded by avoided transportation costs. If railroads expand other activities whose social marginal return exceeds cost, aggregate gains can be much larger.
- **Beyond land-value capitalization.** Donaldson and Hornbeck (2016) quantify market-access gains through land values. Hornbeck and Rotemberg add a distinct surplus channel: output value minus input cost can rise without being fully capitalized into land.
- **Measurement contribution.** The authors digitize county-by-industry Census of Manufactures data for 1860, 1870, and 1880, allowing county-specific revenue, input costs, production elasticities, and input wedges to be measured.
- **Conceptual contribution.** The paper integrates a market-access framework with modern misallocation/productivity accounting. This connects economic history, quantitative geography, and distorted-economy growth accounting.
- **Empirical contribution.** It decomposes local productivity effects into TFPR and AE, showing that railroad-induced productivity growth mostly reflects allocative efficiency rather than technical efficiency.
- **Quantitative contribution.** It embeds the reduced-form estimates in a general equilibrium model with input wedges and Domar aggregation, showing how local shocks translate into national aggregate productivity losses.
- **General lesson.** The paper reframes the value of infrastructure: a technology can be more valuable in an imperfect economy than in an efficient one because it relaxes a constraint on activities that are inefficiently small.

3 Data construction and measurement

3.1 Historical manufacturing data and county-industry concordance

- The central data source is newly digitized county-by-industry tabulations from the U.S. Census of Manufactures for 1860, 1870, and 1880.
- The authors observe factory-gate revenue and expenditure on major inputs: materials, labor, and capital.
- Industries are harmonized across decades. The paper groups the detailed industry labels into 31 main industry groups and also reports checks with more detailed 193 industry categories.
- The main county panel includes 1,802 counties observed in 1860, 1870, and 1880 that report positive nonzero manufacturing activity.
- The data allow the authors to measure not only output and input growth but also county-specific input shares and county-specific production-function elasticities.

Interpretation: The data are unusually rich for nineteenth-century U.S. economic history because they let the authors calculate county-level productivity and allocative efficiency rather than merely observing population, land values, or railroad access. This is what makes the paper able to distinguish *how much* activity expanded from *whether that expansion raised productivity*.

3.2 County aggregate productivity

For county c, the paper defines county aggregate productivity as surplus:

$$Pr_c = R_c - \sum_k E_c^k,$$

where R_c is manufacturing revenue and E_c^k is expenditure on input $k \in \{K, L, M\}$: capital, labor, and materials.

- Revenue is measured at factory gate.
- Input expenditure is the sum of expenditure on capital, labor, and materials.
- County productivity is therefore the dollar surplus generated by manufacturing activity in the county.
- National aggregate productivity is the sum of county productivity.

Interpretation: This definition intentionally focuses on the gap between output value and input costs. If a county expands production and revenues rise more than input costs, productivity rises even if technical efficiency does not change.

3.3 TFPR and allocative efficiency decomposition

The impact of market access on county productivity can be decomposed as:

$$\frac{\partial \ln Pr_c}{\partial \ln MA_c} = n_c \left[\frac{\partial \ln R_c}{\partial \ln MA_c} - \sum_k \alpha_c^k \frac{\partial \ln E_c^k}{\partial \ln MA_c} \right] + n_c \sum_k (\alpha_c^k - s_c^k) \frac{\partial \ln E_c^k}{\partial \ln MA_c}.$$

The first term is TFPR growth. The second term is AE growth.

- $n_c = R_c / Pr_c$ rescales changes in revenue into changes in county productivity.
- $s_c^k = E_c^k / R_c$ is the revenue share of input k.

- α_c^k is the production elasticity of input k.
- $\alpha_c^k - s_c^k$ is the input gap. A positive gap means the value marginal product of input k exceeds its marginal cost.

Interpretation: TFPR asks whether the county gets more revenue from a given bundle of inputs. AE asks whether market access expands input use in counties or inputs where the marginal product exceeds cost. The paper's key finding is that the railroad effect is mostly AE.

3.4 Input wedges and distortions

The paper interprets the difference between production elasticities and expenditure shares as an input wedge:

$$\alpha_c^k > s_c^k \quad \Longleftrightarrow \quad \text{input } k \text{ is inefficiently underused.}$$

- A positive wedge could reflect market power, borrowing constraints, contracting frictions, or other distortions.
- The paper does not need to identify the exact source of the wedge for the aggregate productivity accounting.
- The relevant feature is that production expands in places where input marginal products exceed input costs.

Interpretation: The paper is careful not to equate wedges only with markups. The same productivity logic applies if the distortion comes from input constraints rather than output market power.

3.5 Transportation network and market access data

- The transportation network includes natural waterways, constructed canals, lakes, oceans, and railroads.
- Railroads are added by decade: 1860, 1870, and 1880 in the reduced-form analysis; 1890 in the counterfactual model.
- Freight costs are calculated between county pairs using the cheapest routes in the network.
- Rail and water transport are low-cost; wagon transport is much more expensive; transshipment costs apply when goods switch modes.
- County market access summarizes how cheaply a county can reach all destination markets, weighted by market size.

The first-order market access expression is:

$$MA_o = \sum_d \tau_{od}^{-\nu} L_d,$$

where τ_{od} is trade cost between origin o and destination d, L_d is destination market size, and ν is the trade elasticity.

Interpretation: Counties gain market access when railroads lower their trade costs to many destinations. Counties with good water access sometimes gain less because they already had low-cost routes before railroads expanded.

3.6 Agriculture, population, and counterfactual inputs

- For counterfactuals, the paper extends the analysis beyond manufacturing to the broader county economy.
- Agriculture data come from the Census of Agriculture; population data come from the Census of Population.
- Because detailed input wedges are available only in manufacturing, the baseline assumes county wedges measured in manufacturing also apply to agriculture and other sectors.
- Robustness checks relax this assumption by imposing efficient agriculture or alternative input wedge assignments.
- The counterfactual sample includes 2,722 counties with positive population and positive revenue in agriculture and/or manufacturing in 1890.

Interpretation: This step is necessary because railroads affected the whole economy. The manufacturing data identify wedges and reduced-form impacts, but national aggregate productivity requires modeling the economy-wide allocation of labor, capital, land, and materials.

4 Model and empirical specifications

4.1 Reduced-form county panel specification

The baseline regression is:

$$Y_{ct} = \beta \ln(MA_{ct}) + \gamma_c + \gamma_{st} + g_t(y_c) + g_t(x_c) + \varepsilon_{ct}.$$

- Y_{ct} is an outcome such as log revenue, input expenditure, productivity, TFPR, or AE.
- γ_c are county fixed effects.
- γ_{st} are state-by-year fixed effects.
- $g_t(y_c)$ and $g_t(x_c)$ are year-interacted cubic polynomials in county latitude and longitude.
- Standard errors are clustered by state.
- Coefficients are reported as the effect of a one-standard-deviation greater change in market access from 1860 to 1880.

Interpretation: The specification compares counties within the same state that experience different changes in market access, while allowing flexible geographic trends. The design is not simply "railroad arrives in county"; it uses variation in access generated by the full transportation network.

4.2 County productivity accounting

The paper defines log productivity in a way that uses the revenue and input expenditure components:

$$Y_{ct}^{Prod} = n_c \left[\ln R_{ct} - \sum_k s_c^k \ln E_{ct}^k \right].$$

TFPR and AE are:

$$TFPR_{ct} = n_c \left[\ln R_{ct} - \sum_k \alpha_c^k \ln E_{ct}^k \right], \qquad AE_{ct} = n_c \left[\sum_k (\alpha_c^k - s_c^k) \ln E_{ct}^k \right].$$

Interpretation: These formulas let the same estimated market access effect be separated into technical/revenue productivity and allocative efficiency. If all input gaps were zero, the AE term would disappear.

4.3 General equilibrium counterfactual model

The counterfactual model extends Eaton-Kortum/market-access geography to include input wedges. Market access solves a system in which each county's attractiveness depends on trade costs to all other counties and destination expenditures.

The predicted county productivity effect is:

$$\frac{d \ln Pr_o}{d \ln MA_o} = n_o \sum_k (\alpha_o^k - s_o^k) \frac{d \ln X_o^k}{d \ln MA_o}.$$

National aggregate productivity growth is:

$$APG = \sum_o D_o \sum_k (\alpha_o^k - s_o^k) d \ln X_o^k,$$

where D_o is a Domar weight based on county revenue divided by national value-added.

Interpretation: The counterfactual aggregates local effects using Domar weights because intermediate goods and distortions mean county impacts do not aggregate as simple county shares of final output. This is why the general equilibrium model is essential for national estimates.

4.4 Counterfactual scenarios

- Remove all railroads in 1890.
- Restrict the network to only 1850, 1860, 1870, or 1880 railroads.
- Replace railroads with Fogel's proposed extended canal network.
- Double railroad transportation costs.
- Compare fixed worker utility with fixed total population scenarios.

Interpretation: The counterfactuals separate two margins: where activity locates inside the United States and how much total population/activity the U.S. supports. The fixed-population case isolates reallocation; the fixed-utility case allows the national scale of activity to shrink.

5 Identification strategy and empirical credibility

5.1 Baseline identifying variation

The baseline identifies the effect of market access from changes in relative market access across counties from 1860 to 1880. Counties that become better connected to markets are compared with counties in the same state whose market access changes less, after controlling flexibly for latitude and longitude trends.

Interpretation: The assumption is that absent railroad-driven changes in market access, counties with larger market access growth would have followed similar productivity trends as nearby counties within the same state.

5.2 Why market access is not equivalent to local railroad construction

- A county's market access depends on the entire national transportation network, not only tracks in the county.
- Distant railroads can improve a county's access to consumers and input suppliers.
- Railroads can substitute for or complement water routes, generating variation even where local construction is limited.
- This helps address the concern that local railroad lines were built in counties expected to grow.

Interpretation: The paper's advantage is that the treatment is a network-derived access measure. It is less directly tied to local investment decisions than a simple dummy for whether a county received track.

5.3 Controls for endogenous local railroad construction

Table 6 adds increasingly flexible controls for railroads in the county and nearby buffers:

- any local railroad;
- railroad length;
- cubic polynomial in railroad length;
- railroads within 10 miles;
- railroads within 20, 30, and 40 miles.

Interpretation: If the baseline effect were driven by endogenous local construction, the coefficient on market access should collapse when local and nearby railroads are controlled flexibly. It does not.

5.4 Water-market-access instrument

Counties with high preexisting water market access in 1860 tended to gain less from later railroad expansion because they already had cheap access to markets. The paper uses this as an IV:

$$\Delta \ln MA_{ct} \leftarrow \text{Water market access in 1860} \times \text{decade indicators.}$$

Interpretation: This instrument identifies effects from how the railroad network interacted with older water routes. The exclusion restriction is that counties with high water access would not otherwise have different manufacturing productivity trends except through induced changes in market access.

5.5 Expected canal controls and Borusyak-Hull logic

The paper also controls for expected changes in market access from Fogel's proposed canals that might have been built without railroads. This follows the logic that identification should come from unexpected transportation-network changes rather than from predictable improvements.

Interpretation: This is a falsification/control exercise: if counties on plausible alternative canal routes would have grown for other reasons, controlling for proposed-canal market access should affect the estimates. The estimates remain similar.

5.6 Model validation strategy

- Estimate reduced-form market-access effects using the historical county panel.
- Estimate model parameters using 1890 data and transportation-network information.
- Let only the railroad network vary over time in simulated 1860, 1870, and 1880 data.
- Check whether the model reproduces the reduced-form relationships.
- Use the model for national counterfactuals only after showing that it broadly matches local empirical responses.

Interpretation: The model is not simply calibrated to deliver the desired reduced-form effects. It is disciplined by trade flows, land values, railroad shipments, county data, and measured input wedges.

5.7 Main threats and how the paper addresses them

- **Endogenous railroad placement:** addressed by market access, local railroad controls, water-access IV, and canal counterfactual controls.
- **Pretrends and geography:** addressed with county fixed effects, state-by-year fixed effects, geographic trend polynomials, and appendix robustness.
- **Mis-measured production functions:** addressed with alternative industry groupings and robustness to assumptions on elasticities and wedges.
- **Sector extrapolation:** addressed with robustness assuming efficient agriculture or alternative wedge assignments.
- **Migration assumptions:** addressed by reporting fixed-utility, fixed-population, and intermediate scenarios.

6 Tables and figures

6.1 Figure 1: Cross-county dispersion in expenditure as a share of total revenue

Read: Figure 1 plots county-level total input expenditure as a share of revenue for 1860, 1870, and 1880. The distributions are centered below one, implying that revenue generally exceeds input costs.

Interpretation: The figure visually motivates input wedges. If revenue exceeds input expenditure, then there is scope for productivity gains when production expands. The narrowing of the distribution also suggests that expenditure shares become more concentrated over time, but the main point is the persistent gap below one.

Economic message: The railroads matter because counties are not operating at a point where marginal products equal marginal costs. Expanding production in such counties can raise surplus.

6.2 Figure 2: Waterways and railroads, by decade

Read: Figure 2 maps waterways and the sequential expansion of railroads by 1860, 1870, and 1880. The network becomes denser and extends into the interior.

inefficiently too little usage of input k, which could reflect firms' chosen markups or external factors such as borrowing constraints. Figure 1 shows the distribution of expenditure shares across counties. Average expenditure, as a share of revenue, is 0.8 with a standard deviation of 0.1,

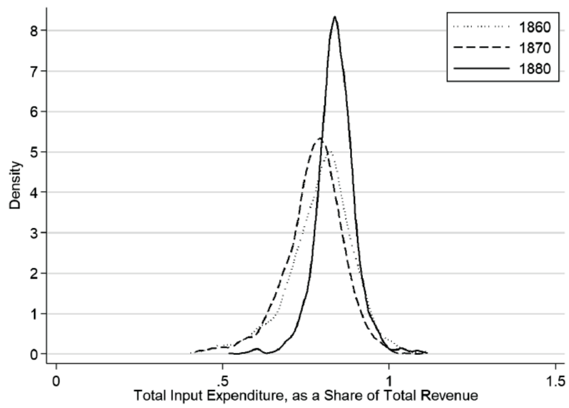

FIG. 1.—Cross-county dispersion in expenditure as a share of total revenue ($\Sigma_k S_i^k$), by decade. Each observation is a county-decade.

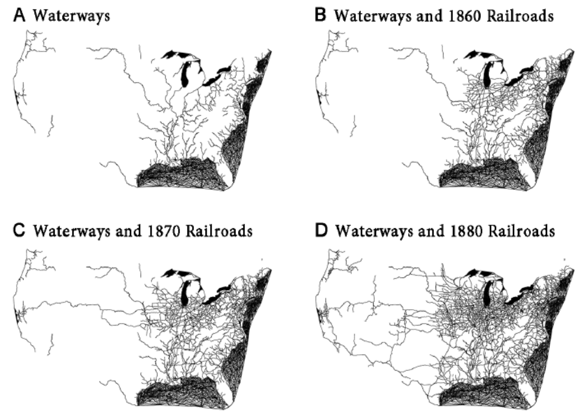

FIG. 2.—Waterways and railroads, by decade. Panel *A* shows the waterway network: natural waterways (including navigable rivers, lakes, and oceans) and constructed canals. Panel *B* adds railroads constructed by 1860, panel *C* adds railroads constructed between 1860 and 1870, and panel *D* adds railroads constructed between 1870 and 1880.

Interpretation: The figure shows that market access changes are not local treatments only. Counties can gain access because distant links close gaps in the national network. Counties already close to water routes are partly substituted away from, while interior counties gain more.

Economic message: The source of variation is the changing topology of a freight network. This motivates the market-access approach rather than a simple local-railroad indicator.

6.3 Figure 3: Calculated changes in log market access, by county

Read: Figure 3 maps changes in log market access from 1860 to 1870 and from 1870 to 1880. Darker counties experience larger increases.

Interpretation: The spatial pattern highlights the interaction between new rail lines and preexisting waterways. Some western and interior counties gain substantially, while counties with strong water access may gain less.

Economic message: The figure previews the empirical comparison: counties differ in how much the national transportation revolution changed their access to markets.

6.4 Table 1: Impacts of market access on county revenue, input expenditure, and productivity

Read: A one-standard-deviation increase in market access raises log revenue by 0.192, capital expenditure by 0.158, labor expenditure by 0.196, materials expenditure by 0.183, and log productivity by 0.204 in the baseline.

Interpretation: The coefficients show a broad expansion of manufacturing activity, not merely a rise in output with fixed inputs. Since productivity rises roughly 20 percent, revenues rise more than costs in surplus

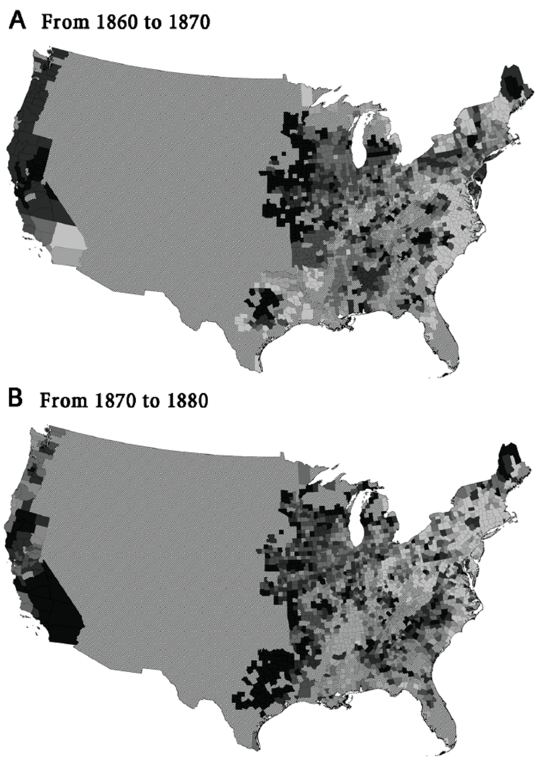

Fig. 3.—Calculated changes in log market access, by county. In each panel, counties are shaded according to their calculated change in market access from 1860 to 1870 (*A*) and from 1870 to 1880 (*B*). Counties are divided into seven groups (with an equal number of counties per group), and darker shades denote larger increases in market access. These maps include the 1,802 sample counties in the regression analysis, which are all counties that report nonzero manufacturing activity from 1860, 1870, and 1880. The excluded geographic areas are crosshatched. County boundaries correspond to county boundaries in 1890.

TABLE 1
Impacts of Market Access on County Revenue, Input Expenditure, and Productivity

	Baseline Specification (1)	Holding Population Fixed at 1860 Levels (2)	County-Level Data Only: 1860–1900 (3)	County-Level Data Only: 1850–1900 (4)
A. Log revenue:				
Log market access	.192	.185	.257	.235
	(.049)	(.047)	(.061)	(.056)
B. Log capital expenditure:				
Log market access	.158	.152	.225	.208
	(.053)	(.051)	(.060)	(.055)
C. Log labor expenditure:				
Log market access	.196	.187	.292	
	(.061)	(.059)	(.068)	
D. Log materials expenditure:				
Log market access	.183	.176	.242	
	(.050)	(.048)	(.062)	
E. Log productivity:				
Log market access	.204	.196	.279	
	(.051)	(.049)	(.057)	
Number of counties	1,802	1,802	1,802	1,437
County-year observations	5,406	5,406	9,010	8,622

Note.—Column 1 reports estimates from eq. (4): for the indicated outcome variable in each panel, we report the estimated impacts of log market access (as defined in eq. [1]), controlling for county fixed effects, state-by-year fixed effects, and year-interacted cubic polynomials in county latitude and longitude. The sample is a balanced panel of 1,802 counties in cols. 1–3 and a balanced panel of 1,437 counties in col. 4. The outcome variables are: the log of total county manufacturing annual revenue (panel A); the log of total county manufacturing annual expenditures on capital, labor, and materials (panels B, C, D); and the log of total county manufacturing revenue minus the weighted logs of total county manufacturing expenditures on capital, labor, and materials (where those weights are the county's average revenue share for that input and the variable is scaled by the ratio of average county revenue to average county productivity, as defined in eq. [2]). In each column, we report the estimated impact of a 1 standard deviation greater change in market access from 1860 to 1880 (e.g., the coefficient in col. 1, panel E, can be interpreted as a relative productivity increase of 20.4% for counties with a 1 standard deviation greater change in market access from 1860 to 1880). This 1 standard deviation greater increase in market access corresponds to a 24% greater increase in market access from 1860 to 1880 for our baseline definition of market access. In col. 2, we calculate market access holding county populations fixed at their 1860 levels, so the only changes in market access come from changes in the transportation network. Columns 3 and 4 use county-level data only, rather than county-by-industry data, which affects only the definition of log productivity in panel E. Column 3 reports estimates for the 1860–1900 period, and col. 4 reports estimates for the 1850–1900 period using available data on county revenue and county capital expenditures in 1850. Robust standard errors clustered by state are reported in parentheses.

Table 2 shows that most of the estimated impact of market access on log terms. The results are similar when holding population fixed at 1860 levels and when using county-level data over longer periods.

Economic message: Market integration increases the scale of economic activity and raises productivity because activity expands where input use was inefficiently low.

6.5 Table 2: Impacts on county productivity, decomposed into TFPR and AE

Read: The baseline productivity effect is 0.204. The TFPR component is only 0.036, whereas AE is 0.168. In the 1860-1900 specification, productivity is 0.279 and AE is 0.258.

Interpretation: Most of the productivity gain comes from allocative efficiency: inputs expand in counties with positive gaps. The TFPR estimate is small and statistically weak, meaning the evidence does not primarily point to railroads directly improving technical efficiency.

Economic message: The productivity gains arise because railroads move the economy along distorted margins, not because counties suddenly become more technically efficient.

TABLE 2
Impacts on County Productivity, Decomposed into TFPR and AE

	Baseline Specification (1)	Detailed Industry Groups (2)	County-Level Data Only: 1860–80 (3)	County-Level Data Only: 1860–1900 (4)
A. Log county productivity:				
Log market access	.204	.204	.204	.279
	(.051)	(.051)	(.051)	(.057)
B. County TFPR:				
Log market access	.036	.038	.038	.020
	(.025)	(.025)	(.026)	(.026)
C. County AE:				
Log market access	.168	.166	.166	.258
	(.051)	(.052)	(.054)	(.067)
Number of counties	1,802	1,802	1,802	1,802
County-year observations	5,406	5,406	5,406	9,010

Note.—Column 1, panel A, corresponds to the estimate reported in col. 1, panel E, in table 1. Column 1 reports estimates from eq. (4): for the indicated outcome variable in each panel, we report the estimated impacts of log market access (as defined in eq. [1]), controlling for county fixed effects, state-by-year fixed effects, and year-interacted cubic polynomials in county latitude and longitude. The sample is a balanced panel of 1,802 counties (1860, 1870, 1880). Panel A reports the estimated impacts on log county productivity (as in panel E of table 1), and panels B and C report the impacts on productivity through changes in county TFPR and changes in county AE as defined in eq. (3). In each column, we report the estimated impact of a 1 standard deviation greater change in market access from 1860 to 1880 (e.g., the coefficient in col. 1, panel A, can be interpreted as a relative productivity increase of 20.4% for counties with a 1 standard deviation greater change in market access from 1860 to 1880). The coefficients in panels B and C imply 3.6% county productivity growth through increases in county TFPR and 16.8% county productivity growth through increases in county AE, respectively. Column 2 calculates the outcome variables using county-by-industry data based on 193 industry categories, rather than the 31 industry categories used in col. 1. Columns 3 and 4 calculate the outcome variables using county-level data, rather than county-by-industry data, for the same period from 1860 to 1880 (in col. 3) and an extended period from 1860 through 1900 (in col. 4). Robust standard errors clustered by state are reported in parentheses.

a more detailed industry classification for this calculation and column 3 reports similar estimates when assuming one production function for all manufacturing industries (from 1860 to 1880). Because market access has similar percent impacts on revenue and each input (table 1), this implies little effect on county TFPR when production exhibits constant re-

TABLE 3
Sources of Growth in County AE

	AE by Input (1)	County Input Gap (2)	County Input Wedge (3)	County Input Cost Share (4)	County Standard Deviation of Wedges (5)
A. Capital:					
Log market access	−.004	.001	.022	−.0004	−.015
	(.009)	(.002)	(.036)	(.0006)	(.044)
B. Labor:					
Log market access	.066	−.001	−.048	−.0012	−.022
	(.015)	(.005)	(.068)	(.0034)	(.044)
C. Materials:					
Log market access	.107	.012	−.028	.0016	.032
	(.049)	(.006)	(.040)	(.0038)	(.054)
Number of counties	1,802	1,802	1,802	1,802	1,802
County-year observations	5,406	5,406	5,406	5,406	5,406

Note.—For the indicated outcome variable, each column and panel reports the estimated impact of log market access from our baseline specification (as in col. 1 of table 1). Column 1 reports impacts on county productivity through changes in county allocative efficiency (as in table 2, panel C, col. 1) through changes in capital (panel A), labor (panel B), and materials (panel C). Column 2 reports impacts on county-level input gaps (defined as the input's cost share minus its revenue share in that decade), col. 3 reports impacts on county-level input wedges (defined as the input's cost share divided by its revenue share, minus one, in that decade), and col. 4 reports impacts on county-level cost shares (defined as the national industry-level cost shares in each decade multiplied by the share of county revenue in each industry in that decade). Column 5 reports impacts on counties' standard deviation of input wedges across industries in that county and decade. All regressions include county fixed effects, state-by-year fixed effects, and year-specific cubic polynomials in county latitude and longitude. We continue to report the estimated impact of a 1 standard deviation greater change in market access from 1860 to 1880. The sample is a balanced panel of 1,802 counties (1860, 1870, 1880). Robust standard errors clustered by state are reported in parentheses.

6.6 Table 3: Sources of growth in county AE

Read: AE growth is driven mainly by materials and labor, with materials contributing 0.107 and labor 0.066. Capital contributes little in this table. Columns on input gaps and cost shares show little evidence that wedges themselves change with market access.

Interpretation: Market access increases inputs, but does not appear to reduce the distortions that create the gaps. This supports the paper's modeling assumption that wedges are exogenous to the railroad shock.

Economic message: Railroads raise productivity by expanding activity in distorted locations, not by curing the distortions.

6.7 Table 4: Impacts of market access, county-by-industry-level regressions

Read: At the county-industry level, the pooled productivity coefficient is 0.189, with AE 0.133 and TFPR 0.056. Effects vary by industry group, with stronger productivity effects in lumber/wood and metals/metal products and weak effects in food and beverage.

Interpretation: The county-level result is not an artifact of aggregation alone. The same AE-centered pattern appears within industries. Industry heterogeneity is expected because industries differ in input intensity, market reach, and initial gaps.

Economic message: The mechanism operates through expansion of particular production activities rather than through a uniform aggregate shock.

TABLE 4
IMPACTS OF MARKET ACCESS, COUNTY-BY-INDUSTRY-LEVEL REGRESSIONS

			By Industry Group			
	Pooled Specification (1)	Weighted by 1860 Revenue Share (2)	Clothing, Textiles, Leather (3)	Food and Beverage (4)	Lumber and Wood Products (5)	Metals and Metal Products (6)
A. Log productivity:						
Log market access	.189	.160	.186	.007	.220	.303
	(.058)	(.052)	(.096)	(.056)	(.149)	(.159)
B. TFPR:						
Log market access	.056	.044	.059	−.012	.075	.054
	(.023)	(.025)	(.073)	(.023)	(.038)	(.123)
C. AE:						
Log market access	.133	.116	.127	.019	.145	.249
	(.058)	(.043)	(.089)	(.063)	(.139)	(.119)
Number of counties	1,800	1,800	994	1,338	1,480	709
County-year observations	5,400	5,400	2,640	3,665	3,984	1,860

NOTE.—This table reports estimates from regressions at the county-by-industry level, after aggregating the more detailed industries to five industry groups: clothing, textiles, leather; food and beverage; lumber and wood products; metals and metal products; and other industries. We extend our baseline estimating eq. (4) to include county-industry fixed effects and state-year-industry fixed effects. The sample is drawn from our main balanced panel of 1,802 counties in 1860, 1870, and 1880, though each industry group is not reported in each county and decade. We omit county-industries that appear only once but do not restrict the sample to county-industries that appear all 3 years. We continue to report the estimated impact of a 1 standard deviation greater change in market access from 1860 to 1880. Robust standard errors clustered by state are reported in parentheses. Column 1 reports estimated average impacts of market access on county-industry productivity, county-industry TFPR, and county-industry AE. Column 2 reports estimates when weighting county-industries by their 1860 share of county revenue. Columns 3–6 report industry-specific effects of market access from separate regressions for each consistent aggregated industry group.

TABLE 5
IMPACTS OF MARKET ACCESS ON COUNTY INDUSTRIES, ESTABLISHMENTS, AND SECTOR SHARES

		Log Average Establishment Size			County Manufacturing Share of:			
	Log Number of Industries (1)	Revenue/ Establishment (2)	Workers/ Establishment (3)	Log Number of Establishments (4)	Revenue (5)	Value-Added (6)	Surplus (7)	Employment (8)
ket access	.028	.021	.024	.172	.0076	.0001	−.0019	.0045
	(.021)	(.030)	(.043)	(.034)	(.0078)	(.0059)	(.0090)	(.0048)
of counties	1,802	1,802	1,802	1,802	1,774	1,774	1,713	1,687
year observations	5,406	5,406	5,406	5,406	5,322	5,322	5,139	5,061

—For the indicated outcome variable, each column reports the estimated impact of log market access from our baseline specification (as in col. 1 of In col. 1, the outcome variable is the log number of manufacturing industries reporting positive output in the county. In cols. 2 and 3, the outcome are log average manufacturing establishment size in the county, based on revenue per establishment (col. 2) or workers per establishment (col. 3). In e outcome variable is the log number of manufacturing establishments in the county. In cols. 5–8, the outcome variables are the county's manufac- are of total values for manufacturing and agriculture: revenue (col. 5); value-added (col. 6), which for manufacturing is defined as revenue minus s expenditures and for agriculture is defined as 92% of revenue; surplus (col. 7), which for manufacturing is defined as revenue minus all input ex- es and for agriculture is defined as the value of land multiplied by the state mortgage interest rate; and employment (col. 8). All regressions include xed effects, state-by-year fixed effects, and year-interacted cubic polynomials in county latitude and longitude. The samples are drawn from our main l panel of 1,802 counties in 1860, 1870, and 1880, which for cols. 5–8 is smaller due to missing data for some counties in some years. We continue to e estimated impact of a 1 standard deviation greater change in market access from 1860 to 1880 in the full sample of 1,802 counties. Robust standard ustered by state are reported in parentheses.

6.8 Table 5: Impacts of market access on county industries, establishments, and sector shares

Read: Market access raises the number of establishments by 0.172 but has smaller and less precise effects on the number of industries and average establishment size. It has little effect on the manufacturing share of revenue, value-added, surplus, or employment.

Interpretation: The growth is extensive within manufacturing: more establishments, not necessarily larger establishments. The lack of large sector-share effects means the paper is not mainly documenting a shift from agriculture to manufacturing.

Economic message: Railroads expand economic activity broadly rather than simply reallocating a county from agriculture into manufacturing.

6.9 Table 6: Impacts of market access, controlling flexibly for local railroad construction

Read: The productivity coefficient remains sizable when controls are added for local railroads, railroad length, nonlinear railroad mileage, and nearby railroad buffers. The productivity effect moves from 0.204 to 0.142 in the most saturated specification; AE remains positive at 0.129.

Interpretation: The estimates are not mechanically driven by counties receiving local rail lines. Distant network changes and interactions with the waterway network continue to generate variation in market access.

Economic message: This table is central to identification: the railroad network matters as a market integration system, not only as local construction.

TABLE 6
IMPACTS OF MARKET ACCESS, CONTROLLING FLEXIBLY FOR LOCAL RAILROAD CONSTRUCTION

	(1)	(2)	(3)	(4)	(5)	(6)
A. Log county productivity:						
Log market access	.204	.218	.198	.197	.175	.142
	(.051)	(.056)	(.057)	(.058)	(.058)	(.056)
B. County TFPR:						
Log market access	.036	.049	.047	.041	.032	.013
	(.025)	(.030)	(.030)	(.030)	(.032)	(.035)
C. County AE:						
Log market access	.168	.169	.152	.156	.143	.129
	(.051)	(.055)	(.056)	(.055)	(.055)	(.054)
Additional controls:						
Any railroad	No	Yes	Yes	Yes	Yes	Yes
Railroad length	No	No	Yes	Yes	Yes	Yes
Railroad length polynomial	No	No	No	Yes	Yes	Yes
Railroads in nearby buffer	No	No	No	No	Yes	Yes
Railroads in further buffers	No	No	No	No	No	Yes
Number of counties	1,802	1,802	1,802	1,802	1,802	1,802
County-year observations	5,406	5,406	5,406	5,406	5,406	5,406

NOTE.—Column 1 reports the estimated impact of market access from the baseline specification (as in col. 1 of table 2). Column 2 includes an additional control for whether a county contains any railroad track. Column 3 also controls for the length of railroad track in the county, and col. 4 controls for a cubic polynomial function of the railroad track mileage in a county. Column 5 includes additional controls for whether a county contains any railroad track within 10 miles of the county boundary and a cubic polynomial function of the railroad track mileage within 10 miles of the county boundary. Column 6 adds controls for separate cubic polynomial functions of railroad track within 20 miles, within 30 miles, and within 40 miles of the county. All regressions include county fixed effects, state-by-year fixed effects, and year-specific cubic polynomials in county latitude and longitude. We continue to report the estimated impact of a 1 standard deviation greater change in market access from 1860 to 1880. The sample is a balanced panel of 1,802 counties (1860, 1870, 1880). Robust standard errors clustered by state are reported in parentheses.

Borusyak and Hull (2023) suggest controlling for the "expected" change in market access when analyzing the actual change, given other potential changes in the transportation network, so identification comes

TABLE 7
IMPACTS OF MARKET ACCESS, USING VARIATION IN WATER MARKET ACCESS

	LOG MARKET ACCESS	2SLS			CONTROLLING FOR PROPOSED CANALS		
	OLS (1)	County Productivity (2)	TFPR (3)	AE (4)	County Productivity (5)	TFPR (6)	AE (7)
rket access		.221	.040	.193	.203	.036	.167
		(.029, .435)	(−.111, .315)	(.001, .367)	(.051)	(.026)	(.051)
ents:							
ater market access in 1860 × year = 1870	−1.003						
	(.212)						
ater market access in 1860 × year = 1880	−1.780						
	(.245)						
gen-Paap *F*-statistic		27.2	27.2	27.2			
of counties	1,802	1,802	1,802	1,802	1,802	1,802	1,802
year observations	5,406	5,406	5,406	5,406	5,406	5,406	5,406

.—Column 1 reports the impact of log water market access in 1860 on changes in log market access from 1860 to 1870 and changes in log market om 1870 to 1880: log market access is regressed on log water market access in 1860, interacted with year fixed effects for 1870 and 1880. Column 2 the estimated impact of log market access on county productivity, instrumenting for log market access using the first-stage relationships reported . Columns 3 and 4 report corresponding 2SLS estimates for county TFPR and county AE. For the 2SLS specifications, we report the Andrews and Sun (2018) two-step weak-instrument-robust confidence sets, as well as the first-stage *F*-statistic. For cols. 4–6, as an additional control to line OLS specification we include the market access that counties would have had if the railroad network had never been built but instead Fogel's proposed counterfactual canal network had been constructed. Robust standard errors clustered by state are reported in parentheses. All regres- clude county fixed effects, state-by-year fixed effects, and year-interacted cubic polynomials in county latitude and longitude. In cols. 2–7, we con- report the estimated impact of a 1 standard deviation greater change in market access from 1860 to 1880. The sample is a balanced panel of 1,802 s (1860, 1870, 1880).

6.10 Table 7: Impacts of market access, using variation in water market access

Read: High 1860 water market access predicts smaller increases in later market access. IV estimates using water market access produce productivity, TFPR, and AE effects similar in magnitude to OLS, though less precise. Specifications controlling for proposed canals also remain similar.

Interpretation: The table uses a historical substitute for railroads - water routes - to isolate plausibly exogenous variation. Counties that already had cheap water access gained less from rail expansion, creating a first-stage for market access.

Economic message: The results survive an identification strategy based on the interaction between new rail technology and inherited transportation geography.

6.11 Figure 4: Counterfactual changes in market access, by county

Read: Figure 4 maps the decline in market access if the 1890 railroad network were removed. Darker counties experience larger market-access losses.

Interpretation: The largest losses concentrate in places that depended on railroads to overcome distance from waterways. The map shows how the no-railroads counterfactual shifts economic opportunity back toward water-connected regions.

Economic message: Removing railroads would not simply raise freight costs uniformly. It would reshape the geography of market integration.

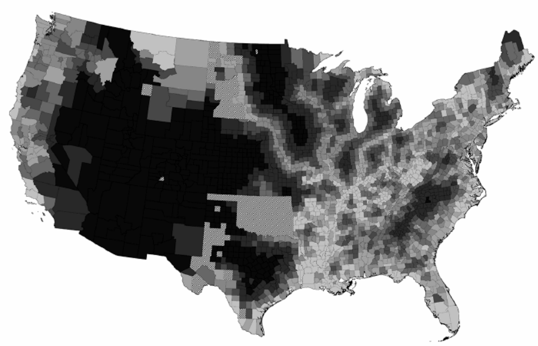

Fig. 4.—Counterfactual changes in market access, by county. This map shows counties shaded according to their change in market access from 1890 to the baseline counterfactual scenario without railroads and where population is allowed to decline: darker shades denote larger declines in market access, and counties are divided into seven equal groups. This counterfactual sample includes all 2,722 counties that report positive population and

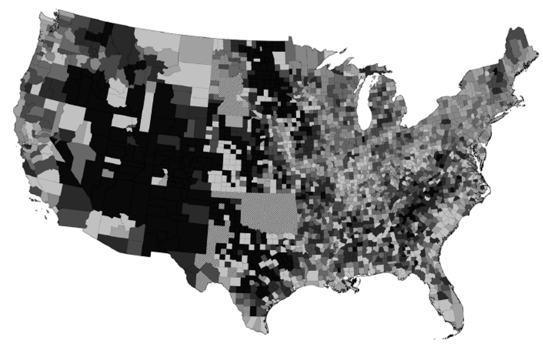

Fig. 5.—Counterfactual changes in productivity, by county. This map shows counties shaded according to their change in productivity from 1890 to the baseline counterfactual scenario without railroads and where population is allowed to decline: darker shades denote larger declines in productivity, and counties are divided into seven equal groups. This counterfactual sample includes all 2,722 counties that report positive population and pos-

6.12 Figure 5: Counterfactual changes in productivity, by county

Read: Figure 5 maps county productivity declines in the no-railroads counterfactual. Darker counties experience larger productivity losses.

Interpretation: Productivity losses combine market-access declines with local input gaps. A county can lose more productivity if it both loses access and has large positive wedges that made railroad-induced expansion valuable.

Economic message: The map is the spatial version of the paper's central mechanism: infrastructure affects aggregate productivity when it expands activity in distorted places.

6.13 Table 8: Impacts of market access on model-implied values

Read: The model-implied nominal productivity effect is 0.197, close to the reduced-form 0.204. The model-implied real productivity effect is 0.257. Real revenue, labor, and materials responses are larger than nominal responses.

Interpretation: The model validates the reduced-form pattern even though it is not fit to match these exact coefficients. The real effect exceeds the nominal effect because market access lowers input and output prices.

Economic message: The quantitative model is credible because it reproduces the key empirical relationship and shows that nominal regressions may understate real allocative gains.

TABLE 8
IMPACTS OF MARKET ACCESS ON MODEL-IMPLIED VALUES

		MODEL-IMPLIED VALUES	
	BASELINE SPECIFICATION (1)	Nominal (2)	Real (3)
A. Log revenue:			
Log market access	.192	.259	.333
	(.049)		
B. Log capital expenditure:			
Log market access	.158	.259	.259
	(.053)		
C. Log labor expenditure:			
Log market access	.196	.259	.337
	(.061)		
D. Log materials expenditure:			
Log market access	.183	.259	.337
	(.050)		
E. Log productivity:			
Log market access	.204	.197	.257
	(.051)		
Number of counties	1,802	1,802	1,802
County-year observations	5,406	5,406	5,406

NOTE.—Column 1 reports estimates from col. 1 of table 1: for the indicated outcome variable in each panel, we report the estimated impacts of log market access (as defined in eq. [1]), controlling for county fixed effects, state-by-year fixed effects, and year-interacted cubic polynomials in county latitude and longitude. The sample is a balanced panel of 1,802 counties (1860, 1870, 1880). We report the estimated impact of a 1 standard deviation greater change in market access from 1860 to 1880. Robust standard errors clustered by state are reported in parentheses. Columns 2 and 3 show the relationship between log market access and model-predicted values in 1860, 1870, and 1880, where the only primitive allowed to vary over "time" is the railroad network and the model parameters are estimated in 1890. Column 2 reports impacts on nominal values for the outcome variables, and col. 3 reports impacts on real values for the outcome variables.

replicates the reduced-form results despite our not disciplining model parameters with the estimated relationship between market access and manufacturing revenues or expenditures. This similarity reflects two countervailing forces. The model reflects a somewhat larger response of real

TABLE 9
COUNTERFACTUAL IMPACTS ON NATIONAL AGGREGATE PRODUCTIVITY

		RESTRICTED RAILROAD NETWORKS					
	BASELINE: NO RAILROADS (1)	Only 1850 Railroads (2)	Only 1860 Railroads (3)	Only 1870 Railroads (4)	Only 1880 Railroads (5)	NO RAILROADS, EXTENDED CANALS (6)	ALL RAILROADS, TWICE THE COST (7)
rfactual scenarios:							
olding worker utility constant:							
ange in aggregate productivity (%)	−26.7	−22.3	−15.6	−9.9	−2.6	−23.5	−8.7
olding total population constant:							
ange in aggregate productivity (%)	−5.5	−5.0	−4.1	−2.7	−.7	−4.4	−1.4
ange in utility (%)	−32.7	−27.2	−18.3	−11.4	−2.9	−29.0	−11.1

.—Each column reports the estimated change in national aggregate productivity from counterfactual changes in the transportation network. reports estimates from our baseline scenario, which holds worker utility constant in the counterfactual and allows for declines in total population. reports estimates from an alternative scenario, which holds total population fixed, and so we also report the associated decline in worker utility. In arios, population is allowed to relocate endogenously within the country. The sample includes all 2,722 counties that report positive population itive revenue (agriculture and/or manufacturing) in 1890. Column 1 reports impacts under our baseline counterfactual scenario, which removes oads in 1890. Columns 2–5 report impacts under more moderate counterfactual scenarios, which restrict the railroad network to those railroads l been constructed by 1850 (col. 2), by 1860 (col. 3), by 1870 (col. 4), or by 1880 (col. 5). Column 6 reports impacts from replacing the railroads sible extensions to the canal network, as proposed by Fogel (1964). Column 7 reports impacts from maintaining the 1890 railroad network but g the cost of transportation along all railroads.

6.14 Table 9: Counterfactual impacts on national aggregate productivity

Read: Removing all railroads lowers aggregate productivity by 26.7 percent under fixed worker utility and by 5.5 percent under fixed total population. Replacing the network with only earlier railroad networks also produces large losses. Extended canals mitigate only a small part of the loss, while doubling railroad costs still leaves major gains relative to no railroads.

Interpretation: The fixed-utility scenario lets population decline and captures the scale effect of railroads on U.S. economic activity. The fixed-population scenario isolates reallocative losses. Both are meaningful: the first reflects nineteenth-century migration margins, while the second shows that even with the same population, spatial reallocation losses are large.

Economic message: The railroads' value comes partly from increasing aggregate scale and partly from reallocating activity toward places where marginal products exceeded costs.

6.15 Table 10: Counterfactual impacts on national aggregate productivity, robustness

Read: The baseline no-railroad loss of 26.7 percent remains large under alternative assumptions: efficient agriculture, alternative capital and labor wedges, efficient capital, alternative trade elasticities, and alternative average transported-good prices.

Interpretation: The estimates are moderately sensitive to some price assumptions but not to the core trade elasticity. Assuming efficient agriculture reduces the loss, but not enough to eliminate the large aggregate effect. Capital wedge concerns also do not drive the result because capital is a small share of expenditure.

Economic message: The central quantitative conclusion is robust: removing railroads sharply reduces aggregate productivity once input distortions are recognized.

TABLE 10
COUNTERFACTUAL IMPACTS ON NATIONAL AGGREGATE PRODUCTIVITY, ROBUSTNESS

	Baseline (1)	Efficient Agriculture (2)	Use Materials Wedge for: Capital (3)	Use Materials Wedge for: Capital and Labor (4)	Efficient Capital (5)	Alternative Trade Elasticities $\theta = 2.0$ (6)	Alternative Trade Elasticities $\theta = 3.9$ (7)	Alternative Trade Elasticities $\theta = 8.2$ (8)	Alternative Average Prices $\bar{P} = 20$ (9)	Alternative Average Prices $\bar{P} = 50$ (10)
worker utility:										
ge in aggregate productivity (%)	−26.7	−16.5	−24.3	−20.2	−23.4	−27.6	−26.1	−23.6	−36.1	−23.2
total population:										
ge in aggregate productivity (%)	−5.5	−4.6	−4.6	−6.1	−4.1	−5.7	−5.4	−4.0	−7.6	−4.7
ge in utility (%)	−32.7	−32.9	−32.7	−33.0	−32.7	−34.1	−31.9	−29.0	−43.5	−28.6

—Each column reports impacts under our baseline counterfactual scenario that removes all railroads in 1890, as in col. 1 of table 9. Panel A reports s from our baseline scenario, which holds worker utility constant in the counterfactual and allows for declines in total population. Panel B reports s from an alternative scenario, which holds total population fixed, and so we also report the associated decline in worker utility. In all scenarios, ion is allowed to relocate endogenously within the country. The sample includes all 2,722 counties that report positive population and positive rev- griculture and/or manufacturing) in 1890. Columns 2–10 report robustness of our baseline estimates (in col. 1) under alternative parameters. Our estimates use our estimated value for θ, the trade elasticity, of 3.05 with a 95% confidence interval between 1.95 and 3.90. In col. 2, we reduce the d degree of input distortions in each county by assuming that the agricultural sector is efficient and apply our estimated manufacturing wedges only unty's manufacturing share of combined output across manufacturing and agriculture. Columns 3 and 4 use the estimated materials wedge in each o assign counties' capital wedge (col. 3) or capital wedge and labor wedge (col. 4). Column 5 assumes that capital use is efficient, such that there is for capital (or a wedge of zero). In cols. 6 and 7, we alternatively impose values for θ of 1.95 or 3.90; in col. 8, we impose a value of 8.22 from on and Hornbeck (2016). Our baseline estimates also use our estimated value for $\bar{P}$ of 38.7, the average price of transported goods, which scales med transportation costs into proportional costs. In cols. 9 and 10, we alternatively impose values for $\bar{P}$ of 20 or 50.

7 Interpretive synthesis

- **Local evidence and aggregate model fit together.** The reduced-form regressions show that market access raises local surplus. The model explains why local surplus effects aggregate into large national productivity gains.
- **AE is the paper's core object.** The paper is less about discovering that railroads increased activity, and more about showing that activity expansion raised productivity because input use was distorted.
- **Infrastructure amplifies distorted margins.** If the economy has positive wedges, a transportation improvement acts like a subsidy to production. That can generate gains far beyond the transportation sector.
- **Distortions need not fall.** The railroads do not reduce markups, borrowing constraints, or other wedges. They make those wedges economically important by increasing the scale of production where gaps are positive.
- **The paper changes the interpretation of historical growth.** Nineteenth-century U.S. railroads did not just move goods more cheaply; they helped determine how much economic activity the country could support and where that activity occurred.

8 Short summary

- The paper studies the aggregate productivity gains from U.S. railroads in a distorted economy.
- It digitizes county-by-industry manufacturing data for 1860, 1870, and 1880.
- County productivity is defined as revenue minus input costs.
- Productivity growth is decomposed into TFPR and AE.
- Market access is calculated using the full transportation network of waterways, canals, railroads, and wagon links.
- A one-standard-deviation increase in market access raises county productivity by about 20 percent.

- Most of this effect is driven by AE, not TFPR.
- The railroads expand input use in counties where inputs were underused relative to their marginal products.
- The effect is robust to local railroad controls and to IV strategies based on water market access.
- The model shows that removing railroads in 1890 would reduce national aggregate productivity by 26.7 percent under fixed utility or 5.5 percent under fixed population.
- These gains are much larger than traditional social-savings estimates because the model allows infrastructure to expand other distorted activities.

9 Conclusion

9.1 Core conclusion

Hornbeck and Rotemberg show that railroads generated large aggregate productivity gains because they expanded economic activity in a distorted economy. The key mechanism is not a direct improvement in county technical efficiency, but an increase in input use where the value marginal product of inputs exceeded their marginal cost.

9.2 Economic intuition

If an economy is efficient, a transportation improvement mainly saves transportation resources and reallocates activity without creating additional surplus outside those savings. If an economy is distorted, however, the same transportation improvement can expand activities that were inefficiently small. Then the value of new output can exceed the cost of new inputs, raising aggregate productivity.

9.3 Incremental contribution

The paper adds a misallocation channel to the classic railroads literature. Fogel's social savings and Donaldson-Hornbeck land-value approaches capture important direct or capitalized gains, but not surplus from expanding distorted activities. This paper shows that ignoring distortions can understate the value of broad infrastructure by a large factor.

9.4 Takeaway

The general lesson is that the social value of infrastructure and general-purpose technologies depends on the surrounding economy. In a distorted economy, new connectivity can generate large indirect productivity gains by encouraging production where social marginal returns are high. With great problems come great possibilities.